Ruitong Tian

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EDUCATION

- **University of Florida** — *2025 – Present*, Gainesville, FL
  Ph.D. student | Sustainable Construction. Advisor: Eric Jing Du.
  Research focus: Dexterous manipulation, robot learning, and tactile sensing.
- **National University of Singapore** — *2023 – 2025*, Singapore
  M.S. & Year 4 Undergraduate Exchange | Infrastructure Digitalization & Management.
  Advisor: Prof. Poh Leong Hien.
- **South China University of Technology** — *2020 – 2024*, Guangzhou, China
  B.E. | Green Intelligent Construction. Advisor: Prof. Nan Hu.

PROJECTS & PUBLICATIONS

ExoBridge: Learning a Bare Hand to Hand-Worn Exoskeleton Mapping through Human Limb Coupling — **Under submission**, *University of Florida*
Ruitong Tian, Xianyao Li, Noah B. Wilson, Fang Xu, Eric Jing Du.

- Developed ExoBridge, a framework for learning exoskeleton motion and tactile signals from bare-hand video.
- Evaluated the learned mapping across four manipulation tasks using over 1,200 demonstrations.

From Wearable Interfaces to Dexterous Policies: Contact Shifts and Tactile Representations — **Under submission**, *University of Florida*
Ruitong Tian, Fang Xu, Noah B. Wilson, Xianyao Li, Eric Jing Du.

- Studied how wearable interfaces and tactile representations shape demonstration-based robot learning.
- Tested Diffusion Policy on four tasks; richer tactile inputs raised pooled success from 35% to 52%.

Semantically Enhanced Eye-Tracking Controls in Human-Robot Interaction: A Comparison Study — **Ergonomics**, **Accepted Oct. 9, 2026**, *University of Florida*
Ruitong Tian, Hengxu You, Fang Xu, Jiahao Wu, Bowen Sun, Thanakon Uthai, Eric Jing Du.

- Studied semantically enhanced gaze control to improve the usability and efficiency of human–robot interaction.
- Co-designed a 30-participant study and analyzed task performance, user experience, and gaze behavior.

Nudge Before You Push: Physics-Aware Navigation via Tactile Probing — **Under submission**, *University of Florida*
Xianyao Li, Fang Xu, **Ruitong Tian**, Bowen Sun, Xiao Hu, Yang Ye, Jing Du.

- Coauthored TANav, a tactile-probing framework for physics-aware robot navigation, evaluated in simulation.

Controlling crack propagation in layered beams with architected lattice-reinforced composite interlayer designs — **Construction and Building Materials (2024)**, *South China University of Technology*
Binglin Xie, **Ruitong Tian**, Haiyang Zhao, Tongtong Ye, Yuan Zhang, Nan Hu.

- Developed architected lattice interlayers for crack control through computational and experimental analysis.

ADDITIONAL EXPERIENCE & AWARDS

Machine Learning for Thermal-Image Damage Prediction — *National University of Singapore*

- Proposed a thermal-image framework for façade crack detection and concrete damage severity prediction.
- **Leadership:** Main Initiator & Vice President, Intelligent Digital Engineering Association (2022–2023); debugged 3D printers and led printing/Cura training.
- **Awards:** Outstanding Undergraduate Thesis (Top 10%); Third Prize, Mathematical Modeling (APMCM, 2023).

TECHNICAL SKILLS

- **Programming:** Python, C++, MATLAB.
- **Machine Learning:** PyTorch, Hugging Face, scikit-learn, imitation learning, Diffusion Policy, CNNs, temporal modeling.
- **Computer Vision & Data:** OpenCV, DINOv2, FastSAM, MediaPipe, NumPy, pandas, SciPy.
- **Robotics & Simulation:** ROS/ROS 2, MuJoCo, Isaac Sim, dexterous manipulation, tactile sensing.
- **Robotic Hardware & Platforms:** DexUMI, XHand, Franka Emika, HoloLens 2.
- **Systems & Computing:** Linux (Ubuntu), Git, CUDA, Slurm, HiPerGator, multi-node GPU training.